Supplemental Material: Exposure to Sexualised Appearance Content on Social Media and Body Satisfaction in Female Adolescents

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1 Preregistration

Table S1. *Preregistration source and confirmatory status of each hypothesis. The original registration labels are given in the third column.*

Test	Exposure measure	Registration (original label)	Status
H1a	Thin & sexualised bodies	Prereg 1 (H1)	Confirmatory, primary
H1b	Beauty filter	Prereg 2 (H1)	Confirmatory, primary
H1c	Beauty label	Prereg 2 (H2)	Confirmatory, secondary
H2a	Thin & sexualised bodies	Prereg 1 (Expl. H1)	Exploratory
H2b	Beauty filter	Prereg 2 (H3)	Confirmatory, secondary
H2c	Beauty label	Prereg 2 (H4)	Confirmatory, secondary
H3a	Thin & sexualised bodies	Prereg 1 (Expl. H2)	Exploratory
H3b	Beauty filter	Prereg 2 (H5)	Confirmatory, secondary
H3c	Beauty label	Prereg 2 (H6)	Confirmatory, secondary
H4a	Thin & sexualised bodies	Prereg 1 (Expl. H3)	Exploratory

2 Preregistration adherence: deviations, unanticipated decisions, and clarifications

This section lists every point at which the study departs from the preregistration (OSF k7953), decides something the preregistration left open, or resolves ambiguous wording. Tables S2 and S3 give the deviations and their reasons, for the statistical analysis and for the classification pipeline. Table S4 gives the decisions on points the preregistration did not cover, and Table S5 the clarifications of ambiguous wording. Rules that were implemented exactly as registered – the exposure windows, zero exposure on days without TikTok use, the content-probability substitution, the imputation model and the one-sided tests – are described in the main text and are not repeated here.

Table S2. *Deviations from the preregistration, with forcing reasons: statistical analysis.*

Preregistered	**Implemented**	**Reason**
Multiple imputation with the pan engine, via mitml.	We used jomoImpute (the jomo engine) with the same imputation model, settings and seed.	Household income, a person-level variable in the registered imputation model, was missing for 28 of 66 participants, and pan cannot impute person-level variables.
A burn-in of 6,000 iterations.	We used 20,000 iterations; everything else was as registered.	A longer burn-in reduces the risk that the retained draws still depend on the chain's starting values; it costs only run time and does not add bias the results.
Gender as a person-level predictor in the imputation model.	Gender was left out.	All participants are female, so gender is a constant.

Preregistered	Implemented	Reason
Exposure analysed in seconds.	Exposure was rescaled to minutes, and coefficients are reported per minute. Models were fitted with the nlminb optimiser. Where nlminb stopped with "singular convergence" because the random-slope variance was essentially zero (the evening beauty-filter and beauty-content models), all 30 imputed datasets were refitted with optim; the morning models were fitted with optim throughout.	These are numerical choices. The model itself is unchanged.

Table S3. *Deviations from the preregistration (continued): content-classification pipeline.*

Preregistered	Implemented	Reason
A fine-tuned vision classifier for sexualised content, trained on previously rated social-media images and selected by macro-F1 on a held-out set.	A zero-shot embedding pre-filter (Marqo-FashionSigLIP) followed by prompt-based classification of clothing, posing and thinness with a vision-language model (Llama 4 Scout), validated with the registered blinded human review.	The VLMs are very expensive to run and we needed to add a pre-filter to enable this analysis on a less extensive pool.

Preregistered	Implemented	Reason
Sexually suggestive posing as the posing component of the sexualisation criterion.	Posing of any kind (Step 5b, stage 1) is the posing component of the confirmatory exposure definition.	Sexually suggestive posing was rare: 11,702 of the 1,050,974 person-crops entering classification (1.1%). Posing for camera of any kind was common (58.9%). An exposure variable that requires suggestive posing would be close to zero on almost every day and could not be tested.
The thinness filter applied before the sexualisation filter.	The order was reversed: clothing, then posing, then thinness.	Body build is the hardest judgement to make from a single still and is most reliable among persons who are revealingly dressed and posing. Because the definition requires all three criteria, the order does not change which stills qualify.
Due to an error, the H1 defiess thin and sexualised female bodies as ‘slim stature and minimal clothing,’ while all the other hypotheses and the preregistered analysis pipeline define it as 'minimal clothing and sexually suggestive body or facial pose'.	We used the definition 'minimal clothing and sexually suggestive body or facial pose' throughout	This was an error in the preregistration.
Iterate on prompts, hyperparameters and thresholds until every step reaches 95% accuracy.	Final accuracies with two blinded reviewers and 1,000 stills per step: human presence 94.4%, female presence 89.5%, clothing pre-filter 78.5%, sexualised presentation 71.3%, thinness 71.5%.	The criterion could not be reached for interpretive classification. Sexualised presentation and thinness are graded judgements, and trained human raters differ among themselves, so near-perfect agreement with a human panel was no longer a realistic target.

Table S4. *Decisions on points the preregistration did not specify or could not have anticipated at the time of planning.*

Point not covered	**Decision**
Donation periods that only partly overlap the EMA window (16 of 924 participant-days without donation data).	These participants were kept. The uncovered days have no derivable exposure and were treated as missing and imputed. The timing of the donation was outside our control.
Donation periods that end before the EMA window (2 participants).	These participants were excluded, because no exposure could be derived for any study day. This follows from inclusion criterion (b).
Slideshow posts (sequences of still images).	The images were used directly as stills. Each image counts as 3 seconds on screen, TikTok's advance rate for photo posts, which we checked in the app. The number of images times 3 seconds is the post's duration (Section 6).bookmark=id.37yailesltw2
Viewing time longer than the content itself (looping videos, slideshows cycling through their images).	Each still counts at most once per watch event, so watching beyond one full pass adds no exposure. The preregistration's rules say which stills were reached, not what happens when they are seen again (Section 6).bookmark=id.37yailesltw2
Watch-duration estimation beyond the two registered rules (drop stills after a switch; the last video in an episode counts as watched in full).	We used a cascade that takes the best available evidence for each watch event (Section 6): the content's own length, bounded by the gap to the next event where that is shorter; donated session boundaries to tell within-session from between-session gaps; the session's end time where a boundary was crossed; the participant's median high-confidence duration as a plausibility bound where no session data existed; and, for the remaining 0.5% of watch events, a random draw from her own high-confidence durations.bookmark=id.37yailesltw2
A session-final video watched by several participants.	The instance with the best-evidenced duration determines which stills are extracted.

Point not covered	Decision
Screening steps needed before classification at scale.	A high-recall clothing pre-filter over 21 garment categories (a person is flagged if the summed revealing probability is at least 0.30 or the highest revealing category at least 0.50) and a person-quality filter (among the three largest persons in the frame; bounding box at least 5% of the frame; at least 20% of the box below the shoulder line; blur score at least 20). Each threshold was set by manual review.
How to judge posing.	In two stages: first whether the person is posing for the camera at all, then whether the posing is sexually suggestive. The first stage defines the pool for the second and gives a variable with a usable base rate.
Person-crops the model cannot judge.	The prompts allow an explicit “indeterminate” answer, which is removed from the qualifying pool as the preregistration asks for clear matches only. Indeterminate answers are counted separately from refusals, and the prompts state the permitted reasons so that the model does not hedge.
Male-presenting persons that pass the deliberately permissive gender filter.	Every prompt instructs the model to answer NO if the person appears male presenting for the reason that we had implemented a high-recall filter earlier in the analysis pipeline.
Detected faces that belong to no detected person.	Faces were matched to persons by bounding-box containment and head position. Unmatched faces (8.2%) were treated as background and excluded.

Table S5. *Clarifications of ambiguous or internally inconsistent preregistration wording.*

Ambiguity in the preregistration	**Resolution**
"Minimal/tight clothing" does not say how tight, or whether one tight garment in an ordinary outfit counts.	A minimal garment is enough on its own. For the tight route the whole visible outfit must be tight; one tight garment with ordinary or loose clothing does not count. This targets an over-inclusion pattern seen in manual review.
Imputation model: "all other EMA variables at the most recent EMA occasion".	"Other" was ambiguous and could mean other than body satisfaction or other than the value being imputed. We read it as other than the value being imputed, so the body-satisfaction rating at the most recent occasion (the lagged item) is included as a predictor.
Tier-2 content-probability substitution: "average probability of the other available videos across the whole 14-day study period", without saying whose videos.	The participant's own proportion across her full donation period, used only on days when none of her videos could be classified.

3 Content extraction pipeline

The computer vision classification pipeline was implemented in Python. Preprocessing and classification steps not requiring a vision-language model (VLM) were performed locally on a Mac Studio M3 Ultra (Python 3.14). All VLM classification (i.e., Llama4 scout) was performed using vLLM [1], a high-throughput inference and serving engine for large language models, on four NVIDIA A100 GPUs in parallel provided by the Cambridge Service for Data Driven Discovery (CSD3), a UK Tier-2 high-performance computing facility.[1] Python 3.11 was used for this stage. All VLM classification jobs used continuous batching and a checkpointed, resumable design.

[1] https://www.hpc.cam.ac.uk

1 · Two content types → one stream of stills

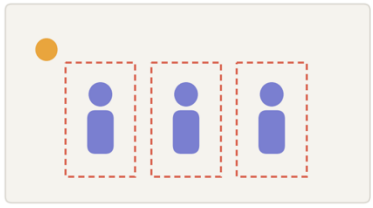


3,283,135 stills on screen · 235,727 videos

3,180,871 video frames · 102,264 slideshow images

on-screen time per still (1 s / 3 s) → exposure seconds

2 · YOLO finds every person on the still



one dashed box per person; box coordinates are fixed here

unit switch: still → persons

5,568,853 persons detected (23.0 per video)

on 3,424,668 stills with a detection record

2,880,586 stills with ≥ 1 person (87.7% on screen)

3 · Visibility check — enough body in view?



pose estimation inside each box: face, torso or thighs visible?

4,710,317 persons pass (84.6%)

858,536 dropped (15.4%)

keypoint confidence ≥ 0.5 on at least one region

4 · Gender — two models look at every person



either model says F → person qualifies (high recall by design)

face-only evidence alone → excluded

3,167,378 female-presenting (56.9% of detected)

274,894 face-only excluded → 2,892,484 qualifying in 1,912,733 stills

the two models agree on 81.5% (n = 3,272,714)

5 · Crops: revealing-clothing prefilter + image quality filter



2,892,470 crops scored

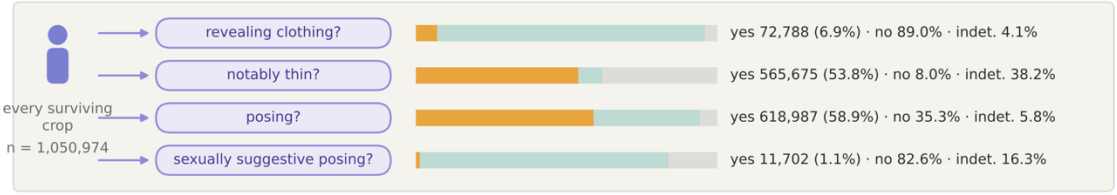
1,590,383 flagged revealing (55.0%)

1,050,974 pass the quality filter (66.1% of revealing)

prefilter favours recall; the VLM re-asks clothing in 6

prefilter: FashionSigLIP, 21 clothing categories · quality: size, sharpness, top-3 agreement, shoulders visible

6 · Scout asks four questions — same pool, every crop



bars: share of the pool answering yes / no / indeterminate (model refusals, ≤ 0.5%, counted with indeterminate)

7 · What the still now carries — one answer per criterion



confirmatory (revealing ∧ posing ∧ thin):

32,441 crops → 30,842 stills (0.94% of stills on screen)

preregistered (∧ suggestive posing): 4,729 crops

Figure S1. *Content extraction pipeline with counts from the final run. Stills on screen (1) are searched for persons (2), who pass a body-visibility check (3) and a two-model gender classification (4); qualifying crops are screened for revealing clothing and image quality (5) and put to four independent questions by the vision-language model (6); a still qualifies when at least one crop meets the exposure definition (7). Panels correspond to Steps 1, 2a, 2b, 3, 4, 5 and 6–7 of the text; panel 6 percentages are of the 1,050,974-crop pool (model refusals, ≤ 0.5%, counted as indeterminate).*

3.1 Step 1 – Still-frame extraction.

TikTok content takes one of two forms: standard video, or slideshows composed of a sequence of static images typically accompanied by music. To enable a unified, frame-based analysis pipeline across both content types, all downloaded content was first converted into a sequence of still images. For video content, one frame per second was extracted using ffmpeg [2], yielding a sequence of still images per video indexed by frame number. Slideshow content required no such extraction, since each constituent image is already a static still; these images were used directly, with no separate processing step needed to convert them into stills. Person detection (Step 2) and all subsequent pipeline steps treat video-derived and slideshow stills identically. The two content types differ only in how long each still was on screen – one second per frame for video, three seconds per image for slideshows – which is recorded per still and enters the exposure computation as a weight, without affecting any step of the extraction or classification pipeline. How that timing was established, how each viewing event's duration was estimated, and how content that could not be classified was handled are set out together in Section 6. In total, 3,283,135 stills were on screen across 235,727 videos (3,180,871 video frames and 102,264 slideshow images).

3.2 Step 2 – Human-presence filter.

Stills were retained only if they clearly contained a person – i.e., a visible face, parts of the torso, or the thighs; stills showing only minor body parts (e.g., a hand within a landscape) were treated as containing no human presence and removed.

3.2.1 Step 2a – Person detection.

We used the Ultralytics YOLO26 model (largest variant, "x") to determine whether at least one person was visible in a given still image [3,4]. Modern YOLO (You Only Look Once) models are trained and benchmarked on COCO (Common Objects in Context), a standard computer vision dataset of more than 200,000 manually annotated images spanning 80 object categories, including people [5]. Detection accuracy on this benchmark is commonly reported as mean average precision (mAP): the average, across all 80 categories, of how accurately each detected bounding box matches the true, human-labelled box. Rather than a single fixed standard for what counts as a correct match, the standard mAP@0.5:0.95 metric averages accuracy across ten standards of increasing strictness, from a lenient one requiring only that the detected and true boxes roughly overlap, up to a strict one requiring the two boxes to align almost exactly – giving a fuller picture of accuracy than any single standard alone. YOLO26 reports an mAP@0.5:0.95 of 40.9 to 57.5 across its five model sizes on COCO, with person detection specifically among the best-performing of the 80 categories, achieving around 90% average precision [6,7]. Person detection with YOLO has also been benchmarked directly against other model families and found to offer the best speed-accuracy trade-off [8]. We used a confidence threshold of 0.25, the Ultralytics default for inference [5]. Detected persons were ranked largest-first by the fraction of the image their bounding box occupied. Person detection was run on 3,424,668 stills from 242,422 videos (including

141,533 stills not on screen within the estimated watch duration, which carry a detection record but do not enter the exposure computation); 2,880,586 stills (84.11%) contained at least one detected person.

3.2.2 Step 2b – Body visibility filter.

For every detected person, the YOLO pose model (largest variant, “x") was applied to a padded crop of the person’s bounding box (10% padding on each side, to avoid cutting off keypoints near the box edge) to detect 17 body landmarks in the standard COCO keypoint format (nose; left/right eye; left/right ear; left/right shoulder; left/right elbow; left/right wrist; left/right hip; left/right knee; left/right ankle). Keypoints were grouped into three body regions relevant to the preregistered visibility criterion: *face* (nose, left eye, right eye, left ear, right ear), *torso* (left shoulder, right shoulder, left hip, right hip), and *thighs* (left hip, right hip, left knee, right knee). A region was considered visible if at least one of its constituent keypoints was detected with a confidence of 0.5 or higher. A person was carried forward only if at least one of the three regions (face, torso, or thighs) was visible by this definition, directly operationalizing the preregistered requirement of a visible face, torso, or thighs. This keypoint-based approach was adopted in preference to a simpler bounding-box-area threshold, since a small but structurally uninformative detection (e.g., a hand or a partial limb within a larger scene) can occupy a bounding box of any size while showing near-zero confidence across all pose keypoints, and would therefore not be reliably filtered by area alone. If the pose model detected more than one set of keypoints within a single padded person-crop (e.g., where padding incidentally included part of a neighboring person), the keypoint set with the highest mean confidence across all 17 landmarks was retained. Keypoint coordinates were converted from crop-relative to original still-image coordinates and stored alongside per-keypoint confidence scores, face/torso/thigh visibility flags, and the combined sufficient-visibility flag, for

use in subsequent pipeline steps (e.g., as the pose-based head-position reference used in the face-to-person matching plausibility check described in Step 3a). Of the 5,568,853 detected persons, 4,710,317 (84.58%) passed this visibility criterion.

3.2.3 Step 2 – Manual validation by independent reviewers.

To validate the human-presence classifier, we conducted a blinded manual validation of 1,000 randomly sampled frames using two independent human reviewers. On average, the reviewers agreed with the classifier's predictions on 94.40% of frames. Interrater reliability between the reviewers was substantial (Cohen's $\kappa = 0.84$; 91.80% agreement).

3.3 Step 3 – Female-presence pre-filter.

The pool was further narrowed to stills depicting a female-presenting person, accepting stills with a clear view of the face and/or torso or thighs. Ambiguous cases where sex could not be reliably assessed were removed, retaining only clear matches for further processing.

3.3.1 Step 3a – Dual-model detection of gender presentation.

For persons carried forward, we applied gender-presentation detection to identify persons classified as female-presenting. Female presentation was inferred using two complementary computer vision models: InsightFace, which derives gender information from facial characteristics and performs particularly well when a clear view of the face is available, and MiVOLO, which incorporates body features and can therefore provide predictions even when the face is partially occluded, turned away, or not fully visible. We combined predictions from both models to increase robustness and to assess agreement between two independent sources of information on visually inferred female presentation. Published benchmarks report gender classification accuracies of

approximately 96–97% for InsightFace-based models [9]. MiVOLO uses the person coordinates identified by YOLO and confirmed by YOLO-pose to classify perceived gender from the whole body. It reports gender classification accuracy of 95.7% on FairFace (a demographically balanced facial attribute benchmark), 96.5–97.4% on Adience (an "in-the-wild" age and gender estimation dataset), and 98.3% on AgeDB (a benchmark for facial age estimation)[10]. Other published studies have used MiVOLO in a similar role – detecting human subjects and classifying gender while discarding low-confidence or person-absent images – supporting its suitability for this filtering task [11–13]. Because InsightFace detects faces on the whole still independently of YOLO's person detections, each InsightFace-detected face was matched to a specific YOLO-detected person to enable a person-level comparison between the two models, rather than relying on an image-level signal alone. A face was matched to a person if its center point fell within that person's bounding box, subject to an additional plausibility check requiring the face to lie within a threshold distance of that person's pose-estimated head position (from YOLO-pose keypoints). This plausibility check was necessary because a bounding box, being a rectangle drawn around an irregular body shape, can incidentally enclose unrelated background content – for example, a photograph or poster depicting a face, visible behind or near a detected person – which a simple containment check cannot distinguish from a face genuinely belonging to that person. Faces that did not fall within any detected person's bounding box, or that failed the head-proximity plausibility check, were treated as not belonging to any detected person (e.g., a background individual) and excluded from further person-level analysis. Across all 5,568,853 detected persons, 3,312,663 (59.49%) were successfully matched to an InsightFace-detected face. To evaluate the consistency of this dual-model approach, we compared MiVOLO and InsightFace gender classifications for the same detected person, restricting the comparison to persons for whom both models produced a

classification: 4,710,317 persons (84.58%) received a MiVOLO classification (i.e., passed the Step 2b visibility filter), 3,312,663 persons (59.49%) had a matched InsightFace face, and 3,272,714 persons (58.77%) had both, forming the comparable sample for this analysis. The two models agreed on person-level gender classification in 2,665,669 of 3,272,714 comparable persons (81.45%). Both models classified the person as female in 51.76% of cases (n = 1,694,047) and as male in 29.69% of cases (n = 971,622). Disagreements were asymmetric: MiVOLO classified a person as female where InsightFace classified the same person as male in 12.00% of comparable persons (n = 392,694), compared to the reverse pattern in 6.55% of persons (n = 214,351) — an approximately 1.8-fold difference. This asymmetry is consistent with the two models' differing input requirements: MiVOLO classifies gender from full-body visual cues and does not depend on a visible face, whereas InsightFace's classification is contingent on successful face detection and may therefore be less reliable for a person whose face is turned away, occluded, at an unusual angle, or otherwise difficult to classify from facial features alone.

Table S6. *Person-level agreement between MiVOLO and InsightFace gender classification, for detected persons classified by both models.*

	Count	% of comparable persons
Both agree — female	1,694,047	51.76%
Both agree — male	971,622	29.69%
Total agreement	**2,665,669**	**81.45%**
Disagree — MiVOLO: female, InsightFace: male	392,694	12.00%
Disagree — MiVOLO: male, InsightFace: female	214,351	6.55%

	Count	% of comparable persons
Total disagreement	**607,045**	**18.55%**

3.3.2 Step 3b – Inclusion criterion.

Operationalizing the preregistered instruction to retain clear matches and remove ambiguous cases, a person was classified as female-presenting if either model returned a female classification for that person, so that a female-presenting person was not missed due to one model's failure to classify her correctly (e.g., an occluded or averted face for InsightFace, or an atypical body pose for MiVOLO). This criterion classified 3,167,378 of 5,568,853 detected persons (56.88%) as female-presenting, corresponding to 2,086,018 stills containing at least one such person.

3.3.3 Step 3c – Face-only exclusion.

Detected female-presenting persons whose only visible body region was the face (i.e., torso and thighs both not visible, per the Step 2b visibility filter) were excluded, directly operationalizing the face-only ambiguous case named in the preregistered criterion above, since body build cannot be assessed when the body is not visible. Excluding face-only persons removed 274,894 of the 3,167,378 female-presenting persons (8.68%), reducing the number of stills containing at least one qualifying female-presenting person from 2,086,018 to 1,912,733 (a reduction of 173,285 stills, 8.31%). A still was retained for Step 5 if it contained at least one non-face-only detected person classified as female-presenting by either model; applying this criterion retained 1,912,733 of the 2,880,586 stills containing at least one detected person (66.40%) and 2,892,484 female-presenting persons for subsequent classification.

3.3.4 Step 3 – Manual validation by independent reviewers.

To validate the gender classifier, we conducted a blinded manual validation of 1,000 randomly sampled frames using two independent human reviewers. On average, the reviewers agreed with the classifier's predictions on 89.45% of frames. Interrater reliability between the reviewers was substantial (Cohen's $\kappa = 0.80$; 89.70% agreement).

3.4 Step 4 – Removal of covered clothing and check of image quality.

This step narrows the retained-person pool to a plausibly revealing subset using a cheap, low-compute pre-filter (4a), then screens that subset for sufficient visual quality (4b), before any expensive vision-language model classification is applied in Step 5.

3.4.1 Step 4a – Embedding-based clothing pre-filter.

The 2,892,484 female-presenting, non-face-only persons retained after Step 3 were screened using a zero-shot classification pre-filter built on Marqo-FashionSigLIP [14], an open-weights (Apache-2.0) multimodal embedding model fine-tuned for fashion product classification. Embedding models encode an image and a set of candidate text labels into a shared vector space and classify by cosine similarity in a single forward pass and is thus much faster than a vision-language model. Across seven public fashion benchmark datasets, Marqo-FashionSigLIP achieves an average text-to-image recall considering only the top retrieved result of 0.121 (average recall across all considered results of 0.231, mean reciprocal rank of 0.239), a category-to-product precision considering only the top retrieved result of 0.706 (average precision across all considered results of 0.687, mean reciprocal rank of 0.744), and a sub-category-to-product precision considering only the top retrieved result of 0.706 (average precision across all considered results of 0.737, mean reciprocal rank of 0.812), compared to FashionCLIP 2.0's corresponding top-result recall of 0.077

and top-result precision values of 0.618 and 0.681, respectively [14]. This step is explicitly a high-recall pre-filter, not a final precise clothing classification: its purpose is to narrow the full retained-person pool down to a plausibly revealing subset for downstream refinement (see Step 5), erring toward inclusion rather than precision. The finalised taxonomy comprised twenty-one candidate garment categories across three groups: nine *exposure* categories capturing skin-revealing garments for which CLIP-family models discriminate reliably (swimwear, lingerie, underwear, fitted top, short shorts, mini skirt, mini dress, low-cut neckline, bodysuit); five *tight-fit* categories capturing body-hugging silhouette independent of skin exposure (bodycon dress, leggings, tight sportswear, sports bra, catsuit), for which CLIP-family discrimination is markedly less reliable and which were therefore tracked as a separate sub-score rather than pooled indiscriminately with the exposure categories; and seven *covered* negative categories (t-shirt, blouse/shirt, sweater/hoodie, jacket/coat, trousers, long/maxi dress, generic fully-clothed), required so the classifier has an explicit basis for contrast rather than trivially assigning high scores to whichever revealing category happens to be least dissimilar. Each category was represented by two to five phrasings (e.g., *swimwear*: "a bikini," "a woman in a bikini," "a two-piece swimsuit," "a one-piece swimsuit," "a person wearing swimwear at the beach or pool"), independently encoded by the text encoder and averaged into a single representative embedding per category:

$$t_c = \frac{1}{k}\sum_{i=1}^{k} \quad TextEncoder(p_i)$$

This prompt ensembling step increases the robustness of each category's embedding without diluting or fragmenting it into multiple competing entries – all phrasings are combined into one vector before any comparison against the image takes place. *Scoring mechanism.* Each retained person-crop was encoded by the image encoder into a single image embedding v, of the same

dimensionality as each category embedding, permitting direct comparison. Cosine similarity was computed between v and every one of the C = 21 category embeddings:

$$s_c = cos(v, t_c) = \frac{v \cdot t_c}{\| v \| \| t_c \|}$$

yielding C similarity values computed independently of one another, each reflecting only how well the image aligns with that single category. These raw similarities were scaled by a temperature factor τ = 100 and passed through a single joint softmax across all C categories simultaneously:

$$p_c = \frac{exp(\tau \cdot s_c)}{\sum_{c\prime=1}^{C} exp(\tau \cdot s_{c\prime})}$$

such that _c=1 p_c = 1 for every person-crop. Raw cosine similarities from CLIP-family embedding spaces are typically clustered closely together regardless of image content; temperature scaling sharpens small genuine differences into a more decisive distribution before the joint softmax is applied. The joint softmax was chosen for interpretability – it yields a single, directly comparable probability distribution over categories per person-crop, from which a well-defined "top category" can be read off – at a known cost: because every category's final score is computed relative to every other category's score, adding or removing categories mechanically redistributes probability mass across the remaining ones, independent of image content. This trade-off, and its consequences for categories with unequal group sizes, is addressed directly in Step 4b. These per-category probabilities were aggregated into three summary statistics:

$$revealing_score = \sum_{c \in R} p_c \quad covered_score = \sum_{c \in C} p_c \quad revealing_max = \max_{c \in R} p_c$$

where R denotes the union of exposure and tight-fit categories (|R| = 14) and C denotes the seven covered categories. A person-crop was flagged *revealing* if either revealing_score ≥ 0.30 or revealing_max ≥ 0.50: the summed criterion catches person-crops whose revealing evidence is distributed across several plausible categories, which a single-category threshold would miss, while the max criterion catches person-crops with one unambiguous, strongly-scoring category even when nothing else contributes meaningfully. Because |R| ≠ |C|, *revealing_score* and *covered_score* are not computed over an equal number of categories; this asymmetry is examined directly in Step 4b. Of the 2,892,470 person-crops classified (14 of the 2,892,484 qualifying persons could not be read from their still), 1,590,383 (54.98%) were flagged as revealing and 1,302,087 (45.02%) as covered. Table S7 shows the distribution of the revealing pool across its fourteen constituent categories, using each person-crop's single highest-scoring revealing category for descriptive labelling only.

Table S7. *Distribution of the revealing pool across its fourteen constituent categories (primary classification run, n = 1,590,383), by each person-crop's highest-scoring revealing category.*

Category	N	% of revealing pool
Fitted top	406,365	25.55%
Catsuit	300,672	18.91%
Low-cut neckline	217,785	13.69%
Short shorts	160,088	10.07%
Swimwear	109,302	6.87%
Sports bra	80,264	5.05%
Mini dress	77,671	4.88%

Category	N	% of revealing pool
Bodysuit	65,287	4.11%
Mini skirt	51,247	3.22%
Tight sportswear	46,458	2.92%
Leggings	20,840	1.31%
Tight bodycon dress	19,396	1.22%
Underwear	18,555	1.17%
Lingerie	16,453	1.03%

3.4.2 Step 4b – Person-quality filter

The 1,590,383 person-crops flagged as revealing by the Step 4a embedding pre-filter were further screened for visual quality before expensive VLM classification. Manual screening of samples from this pool surfaced three key sources of unreliable visual evidence not addressed by the existing visibility criterion: incidental background persons captured in frame alongside the primary subject, framings showing insufficient body extent below the shoulders despite technically satisfying the Step 2b visibility requirement via a single shoulder keypoint, and motion- or focus-blurred person-crops. Four criteria addressing these sources were computed independently against the full pool and combined via logical conjunction, so that all subsequent steps operate on a pool restricted to genuinely assessable person-crops throughout.

Background-person exclusion. Two criteria addressed incidental background persons. First, for stills containing more than three qualifying persons, only the three with the largest bounding-box

area (as a fraction of total image area, bbox_area_frac, already computed at the original YOLO detection stage) were retained; 75,800 person-crops (4.8%) failed this criterion. Second, persons whose bounding box occupied less than 5% of the total frame area (bbox_area_frac < 0.05) were excluded, following manual review of person-crops across the observed range of this measure to establish where a small but genuine subject became indistinguishable from an incidental background figure; 255,209 person-crops (16.1%) failed this criterion.

Shoulders-only geometric exclusion. A keypoint-presence-based visibility flag (i.e., whether any pose keypoint was detected at all below the shoulder line) was judged too coarse to support an exclusion decision on its own, since elbow and wrist keypoints are frequently undetected even when the torso is clearly visible (e.g., arms held close to the body or out of frame). A continuous geometric measure was constructed instead: for each person-crop, the fraction of the YOLO-detected bounding box's height falling below the pose-estimated shoulder line,

$$f = \frac{y_{bbox,bottom} - y_{shoulder}}{y_{bbox,bottom} - y_{bbox,top}}$$

using whichever shoulder keypoint(s) were confidently detected (≥ 0.5; both averaged if both were confident). A threshold of f < 0.20 was adopted following manual review of person-crops across the observed range of f, identifying a boundary below which the visible bounding box extended too little past the shoulders to support further assessment; 293,141 person-crops (18.4%) failed this criterion.

Blur exclusion. Sharpness was assessed using the variance of the Laplacian of each person-crop, computed after Contrast-Limited Adaptive Histogram Equalization (CLAHE, clip limit 2.0, 8×8 tile grid) was applied to correct for a confound identified during manual review: uncorrected Laplacian variance is sensitive to image contrast as well as focus, causing genuinely sharp but

dimly-lit person-crops to be scored as though blurred. Following this correction, all person-crops scoring below 20 were manually reviewed in full and confirmed to be genuinely too dark and/or blurred to assess; a threshold of blur_score ≥ 20 was adopted accordingly. 3,854 person-crops (0.2%) failed this criterion.

Combined criterion. A person-crop was retained for subsequent VLM classification only if it satisfied all four criteria jointly: top-3 rank by area, minimum area (≥ 0.05), sufficient extent below the shoulders (f ≥ 0.20), and sufficient sharpness (blur_score ≥ 20). Of the 1,590,383 person-crops flagged as revealing, 1,050,974 (66.1%) were retained, forming the pool used for all subsequent VLM classification.

Table S8. *Person-quality filter pass rates by criterion, applied to the 1,590,383-person-crop pool retained after Step 4a.*

Criterion	N passing	% passing	N failing	% failing
Top-3 rank by area	1,514,583	95.2%	75,800	4.8%
Area ≥ 0.05	1,335,174	84.0%	255,209	16.1%
Shoulder extent ≥ 0.20	1,297,242	81.6%	293,141	18.4%
Blur score ≥ 20	1,586,529	99.8%	3,854	0.2%
Combined (all four)	**1,050,974**	**66.1%**	**539,409**	**33.9%**

3.4.3 Step 4 – Manual validation by independent reviewers.

We conducted a blinded manual validation of 1,000 randomly sampled frames using two independent human reviewers. On average, the reviewers agreed with the visual model on 78.50% of frames, with substantial interrater reliability between the reviewers (Cohen's $\kappa = 0.65$; 83.5% agreement).

3.5 Step 5 – Sexualised presentation filter.

This step operationalises the first two components of the preregistered sexualisation criterion – minimal/tight clothing and sexually suggestive posing. The two components were classified sequentially rather than in parallel: every person-crop retained after Step 4 was first classified for clothing (5a); person-crops with a positive clothing classification were then classified for deliberate posing (5b), in two stages – posing at all, then, among posing person-crops, sexually suggestive posing specifically. This sequential design reflects both compute economy and measurement logic: the posing judgments are most informative, and their constructs most determinable, within the pool already showing minimal or tight clothing, and each stage's positive pool provides the natural eligible population for the next, progressively stricter judgment. A person-crop is carried forward to Step 6 (thinness) if it satisfies the sexually-suggestive-posing criterion.

3.5.1 Step 5a – VLM clothing classification

Because Marqo-FashionSigLIP is explicitly a high-recall pre-filter, a VLM was used to verify its output, providing an independent, higher-precision classification of the revealing-clothing pool. Several vision-language models were tested during development, including Qwen3-VL-30B-A3B-Instruct (a mixture-of-experts model, approximately 3B active parameters per token) and Qwen3-VL-8B-Instruct, evaluated on a stratified sample of manually reviewed, deliberately difficult person-crops (cases identified in manual review as commonly misclassified, e.g. tight-fitting streetwear combinations and ambiguous neckline depth). Llama 4 Scout (a mixture-of-experts model, 17B active parameters, 109B total) was ultimately selected for all reported VLM classification, run via vLLM across 4 NVIDIA A100 GPUs using tensor parallelism. Every

qualifying person-crop in the Step 4b pool (n = 1,050,974) was individually classified, with no wave-based early exit and no still-level aggregation performed during the run itself. *Prompt development.* Manual rating of stills and prompt testing revealed that the preregistered wording ("minimal/tight clothing, e.g., swimwear, lingerie, tight sportswear") admits two independent, sufficient conditions – minimal or tight – but is itself underspecified regarding how each should be operationalised (e.g., how tight is sufficiently tight, or whether a single tight garment paired with an otherwise ordinary outfit should qualify). An initial prompt permitting standalone tight-fit as a sufficient condition, independent of any skin exposure, was found through manual review to over-include ordinary streetwear styled with a single fitted piece (e.g., a fitted crop top paired with jeans). The final prompt retains standalone tight-fit as a genuine, independent route to a positive classification (consistent with the preregistration's own naming of "tight sportswear" as an example independent of skin exposure, and with the first author's direct knowledge that sexualised tight-fit workout content on this platform frequently shows no skin exposure at all), but requires the tight-fit criterion to hold across the *entire* visible outfit: a tight garment paired with any other visibly ordinary or loose-fitting garment does not qualify. This directly targets the over-inclusion pattern identified in manual review, since a fitted top paired with ordinary jeans – unlike a matching tight top-and-bottom set or a full-body tight garment – fails this joint requirement. A visibility gate and a male-exclusion safeguard (classifying a person-crop as NO if the person appears male, given that the gender presentation filter had been designed to have high recall) were added following the same rationale developed for Step 5b, described below. The final prompt used:

> *"Look at this person's ENTIRE visible outfit as a whole, including any outer layers that may cover part of the outfit. Does this outfit count as minimal or tight, per the criteria below? FIRST, check visibility: only answer 'indeterminate' if the outfit itself genuinely*

cannot be judged in THIS SPECIFIC IMAGE. Common reasons this happens in short-form vertical video include: the torso/lower body is cropped out of the vertical frame, the outfit is blocked by a hand, phone, another person, or an object, or the shot is too dark, blurry, or at an extreme angle to see what the person is wearing clearly. Do not answer 'indeterminate' for any other reason, and do not answer 'indeterminate' merely because you are not fully certain – if the outfit is genuinely visible, answer YES if EITHER: (1) any visible garment is minimal – swimwear, lingerie, underwear, a thong/string-style bottom, a sports bra, or a similar intimate-apparel-style piece, exposing skin; this alone is sufficient for YES regardless of anything else visible in the outfit (e.g., a visible thong worn under a loose t-shirt still counts as YES); OR (2) the visible outfit is tight, closely following the body's shape (e.g., tight leggings, tight shorts, a catsuit, tight sportswear), AND no other visible part of the outfit is ordinary or loose-fitting – if any ordinary/loose garment is visible elsewhere (e.g., tight leggings paired with a loose t-shirt, or a tight top paired with loose trousers), answer NO for this reason. Answer NO for any outfit that is neither minimal (per clause 1) nor entirely tight-fitting throughout what's visible (per clause 2). Also answer NO if the person appears male, regardless of the outfit itself, since this classification applies only to female-presenting persons. Answer with exactly one word: yes, no, or indeterminate."

Prior to the full run, this prompt was validated on a random sample, checking that responses reliably parsed as a single word and that real examples of the two key edge cases (a minimal garment visible alongside a loose covering; a tight garment visible alongside a loose piece elsewhere) were resolved as intended. *Results.* Of the 1,050,974 person-crops classified, 72,788

(6.93%) were classified as wearing a minimal or tight outfit per the criterion above, 934,901 (88.96%) as not, and 42,874 (4.08%) as indeterminate; the model refused to classify 411 (0.04%).

3.5.2 Step 5b – VLM posing classification

All 1,050,974 person-crops in the Step 4b pool were classified for posing in two stages, using the same model and infrastructure as Step 5a, independently of their clothing classification; the three criteria are combined only at Step 6. Stage 1 classifies whether the person is posing for the camera at all; Stage 2, applied to person-crops classified as posing, classifies whether the pose and/or facial expression is sexually suggestive specifically, directly operationalizing the posing component of the preregistered sexualisation criterion ("sexually suggestive posing of body or face, e.g., pouting lips"). The broader Stage 1 judgment serves two roles: it provides a higher-base-rate posing variable for exploratory and sensitivity analysis alongside the stricter preregistered criterion, and it defines the eligible pool for Stage 2, since a person who is not posing at all cannot be posing suggestively.

Stage 1 – posing for camera. The prompt asks whether the person is in a deliberate, composed stance, gesture, or facial expression aimed at the camera, as opposed to being captured naturally while focused on another activity. The prompt was revised from its original early-development wording, adding an explicit clarification that selfie or mirror-selfie framing (including cases where the phone or hand blocks part of the face) is a common, positive example of deliberate posing rather than a visibility problem, plus a male-exclusion safeguard:

> *"Look at this person's pose and facial expression. Is this person POSING for the camera – i.e., in a deliberate, composed stance, gesture, or facial expression aimed at the camera – as opposed to being captured naturally while focused on another activity (e.g.,*

dancing, talking, walking, doing a task) without a deliberately composed pose? Only answer 'indeterminate' if this genuinely cannot be judged in THIS SPECIFIC IMAGE – for example, the person is significantly cropped/cut off, blocked, too dark/blurry, or at an extreme angle such that neither the pose nor the expression can be made out at all, or the image captures an ambiguous mid-motion moment that could equally be a deliberate pose or incidental movement (common when a single frame is pulled from continuous video). A very common example of posing in this content, which should be judged as YES rather than treated as unclear, is a selfie or mirror selfie (typically including parts of the body or the full body), with the camera pointed toward the viewer – whether the phone is held by the person themselves, or set up further away. The phone is sometimes held directly in front of the face, so that the face itself is not visible at all, and sometimes held at an upward or downward angle instead; this framing itself is common and expected in short-form vertical video, and is a positive signal of deliberate posing, not a visibility problem or a reason for doubt – judge based on whatever is visible. Do not answer 'indeterminate' for any other reason, and do not answer 'indeterminate' merely because you are not fully certain – if the pose and/or expression are visible, give your best yes/no answer. Also answer NO if the person appears male, regardless of pose or expression, since this classification applies only to female-presenting persons. Answer with exactly one word: yes, no, or indeterminate."

Results (Stage 1). Of the 1,050,974 person-crops, 618,987 (58.90%) were classified as posing for the camera, 371,075 (35.31%) as not, and 60,912 (5.80%) as indeterminate; no model refusals were observed. *Stage 2: sexually suggestive posing.* An initial implementation judged facial expression and body position as two fully independent sub-judgments (each with its own prompt),

on the reasoning that either channel alone could constitute sufficient evidence and that a joint question would be unable to resolve a framing common to this content: a full-body selfie in which the smartphone itself obscures the face while the body remains clearly visible. Manual review of this design's output found a persistent over-inclusion problem not attributable to either channel specifically: false positives appeared without a clear pattern across face-only, body-only, and both-channel cases, a minority involved a person who was not actually posing in any meaningful sense, some involved a person substantially cropped or otherwise insufficiently visible for a pose to be judged at all, and a small number involved an underlying gender misclassification (a male-presenting person incorrectly retained by Step 3's gender filter). The prompt was revised to a single combined judgment, explicitly separating visibility screening from the substantive suggestiveness judgment, with the visibility check applied as a hard priority before, and independently of, any judgment about suggestiveness. It also explicitly directs that facial expression and body position be weighed together when both are visible (so that two individually milder signals can combine into a clear positive judgment), but that either alone is sufficient when only one is visible (so that partial visibility does not force a weaker or indeterminate outcome), and retains a gender safeguard classifying a person-crop as NO if the person appears male. The final prompt used:

> *"Look at this person's pose and facial expression. Is either of these sexually suggestive? FIRST, check visibility: if the person is significantly cropped/cut off, blocked, too dark/blurry, or at an extreme angle such that neither the pose nor the expression can be made out at all, answer 'indeterminate' – this takes priority over any other consideration below, regardless of any other doubt. A common case in this content is a full-body selfie where the person is holding the camera or phone in front of or near their face. This is NOT by itself a reason to answer 'indeterminate' – the face being partially or fully*

blocked by the phone/hand in this way is common and expected, not a visibility problem; judge based on whatever is visible. If BOTH the face and body are visible, consider them together as a whole – weigh the facial expression and body position in combination, since two milder signals together can add up to a clearly suggestive overall impression even if neither alone would be sufficient on its own. If only ONE of the face or body is visible (the other blocked, out of frame, or otherwise not visible), base your judgment on that one alone – do not answer 'indeterminate' or default to NO merely because only one is available; a clearly suggestive signal in the one visible region is sufficient by itself. FACE signals include: pouting or puckered lips directed at the camera, or lips/mouth held in a clearly deliberate sexualized shape or gesture (e.g., pouting, biting the lip) – an ordinary open-mouth smile or laugh does NOT qualify, even if the lips are visibly parted; or a sustained sultry or smoldering direct gaze at the camera. BODY signals include: an arched back that accentuates the chest or hips, hands positioned to draw attention to the chest, hips, or thighs (e.g., cupping or framing these areas), a bent-over or crouched position angled to emphasize the buttocks toward the camera, or another deliberately sexualized body position. These signals require a deliberate, held body position – a body caught mid-movement or mid-dance that happens to incidentally resemble one of these shapes in a single frame does NOT qualify; the position must appear intentional and composed, not an incidental snapshot of ongoing motion. Answer NO for a clearly visible but ordinary pose or expression, even if confident, cheerful, or camera-facing – e.g., a natural smile, a neutral or happy expression, a standard standing or sitting pose, casual hand gestures, or a typical dance pose. Also answer NO if the person appears male, regardless of pose or expression, since this classification

applies only to female-presenting persons. Answer with exactly one word: yes, no, or indeterminate."

Results (Stage 2). Of the 1,050,974 person-crops, 11,702 (1.11%) were classified as showing sexually suggestive posing, 867,984 (82.59%) as not, and 165,568 (15.75%) as indeterminate; the model refused to classify 5,720 (0.54%). Relative to the full Step 4 pool, sexually suggestive posing was thus a low-base-rate positive classification (11,702 of 1,050,974 person-crops; 1.11%), consistent with the strictness of the preregistered criterion. *Response length.* All Step 5b prompts constrain the model to a maximum output of five tokens, sufficient for the required single-word response but insufficient to produce extended reasoning. This was a deliberate methodological choice: in earlier development, prompt variants that invited the model to state a brief reason alongside an indeterminate judgment appeared to increase the rate of indeterminate responses relative to an otherwise identical prompt without this invitation, suggesting that providing room for justification made the model more willing to hedge rather than improving judgment quality. As in Step 5a and Step 6 (thinness), responses are additionally screened for model refusal (the model declining to engage with the judgment at all), tracked as a distinct *refused* category separate from indeterminate, since a refusal reflects a model limitation rather than evidence about the image.

3.5.3 Step 5 – Manual validation by independent reviewers.

We conducted a blinded manual validation of 1,000 randomly sampled frames using two independent human reviewers. On average, the reviewers agreed with the visual model on 71.30% of frames, with substantial interrater reliability between the reviewers (Cohen's $\kappa = 0.66$; 86.90% agreement).

3.6 Step 6 – Thinness filter.

All 1,050,974 person-crops in the Step 4b pool were classified for a thin body build, independently of the clothing and posing classifications, directly operationalizing the third component of the preregistered sexualisation criterion ("minimal visible body fat"). Thinness was deliberately classified last, and on this pool: piloting showed body build to be the most difficult of the three judgments to determine from a single still, and the pool of person-crops already showing minimal/tight clothing and deliberate posing is the pool in which the visible-body-fat construct is most reliably observable, with the silhouette least obscured by clothing or incidental framing. Using the same model and infrastructure as Step 5, every qualifying person-crop is individually classified, with no wave-based early exit and no still-level aggregation performed during the run itself. *Prompt development.* An initial implementation using only a general visibility-based indeterminate criterion (comparable to the general visibility guidance used in Step 5) produced a substantially elevated indeterminate rate during development. Manual review of a sample of these indeterminate person-crops found that many showed a person whose body was clearly visible but whose clothing – typically loose or baggy garments such as an oversized hoodie or baggy sweatpants – obscured the body's silhouette, a distinct source of ambiguity not addressed by the general visibility criterion, which was oriented toward framing and image-quality problems (cropping, occlusion, blur) rather than clothing coverage. The prompt was revised to name loose/baggy clothing as an explicit, additional indeterminate criterion alongside the existing framing- and quality-based reasons, and to add an explicit instruction that the amount of skin visible is not itself relevant to the judgment – body build should be assessed from the visible silhouette, whether shown through bare skin or through form-fitting clothing, so that a person is not judged indeterminate merely because little skin is shown as long as the body's outline remains

visible. A further revision added camera angle and pose-induced body distortion (e.g., a twisted torso, extreme foreshortening, or a crouched/bent posture) as an additional indeterminate criterion, since these can visually distort apparent body shape independently of framing, image quality, or clothing coverage. Manual review of this design's output when run on the full pool, rather than the narrower, already-filtered pool used during earlier development, surfaced three further issues. First, a number of person-crops received a confident "no" where the body was genuinely obscured by hair or loose clothing rather than the "indeterminate" the visibility criteria were intended to produce. Second, the overall indeterminate rate was substantially elevated relative to earlier development testing; this is at least partly attributable to the broader, less pre-filtered nature of the full pool relative to the narrower pool used earlier, but manual review also identified the model defaulting to indeterminate or a conservative "no" on partial views (e.g., upper chest and shoulders only, common in close-up selfie framing) that were judged by the first author to be genuinely assessable. Third, some fully visible, ordinary slender or fit builds were classified "no," suggesting the model's implicit threshold for "thin" may have required an unusually lean or heavily muscle-defined build rather than an ordinary slender figure. The prompt was revised accordingly: hair was added as an explicitly named occlusion source alongside the existing hand/phone/person examples, with an explicit instruction that an obscured view should be answered "indeterminate" rather than a confident "no"; an explicit clarification was added that a partial view showing only the upper chest, shoulders, and arms can be sufficient evidence on its own; and an explicit clarification was added that an ordinary slender or fit build qualifies as thin, not only an extremely lean or heavily muscle-defined one. Because these are substantive changes to the prompt's decision logic rather than wording or reorganisation alone, this revision was validated on a sample before use at scale. The final prompt used:

"Look at this person's visible body build. Does this person show a thin body build (i.e., minimal visible body fat)? Only answer 'indeterminate' if the body build itself genuinely cannot be judged in THIS SPECIFIC IMAGE. Common reasons this happens in short-form vertical video include: the torso/limbs are cropped out of the vertical frame (common in close-up or face-focused shots), the view is blocked by a hand, phone, hair, or another person (common in selfie-style or handheld footage), the shot is too dark, blurry, or at an extreme angle to see the body clearly, the person's clothing is loose/baggy enough to obscure the body's actual silhouette (e.g., an oversized hoodie, baggy sweatpants, or a loose oversized garment – common in streetwear-style content), or the person's pose or camera angle distorts or compresses their visible body shape in a way that makes build genuinely unclear (e.g., a twisted torso, an extreme foreshortening angle, or a crouched/bent posture). When any of these visibility problems is genuinely present, answer 'indeterminate' – do NOT answer 'no' merely because part of the body is obscured by hair or loose clothing; an obscured view is a reason for 'indeterminate', not a basis for a confident 'no'. A partial view showing only the upper chest, shoulders, and arms (without the full torso or lower body visible) can still be sufficient to judge build – visible collarbone definition, shoulder/arm proportions, and upper-chest shape are valid evidence on their own. Do not default to 'indeterminate' or a conservative 'no' merely because the view is partial or close-up, as long as enough of the visible area shows a clear build. A normal slender or fit build qualifies as thin for this judgment – this does NOT require an extremely lean, visibly underweight, or heavily muscle-defined build; an ordinary slim figure with minimal visible body fat, even if not exceptionally thin, should be answered YES. The amount of skin visible is NOT relevant

to this judgment – base your answer on the visible body SILHOUETTE and shape, whether shown through bare skin or through clothing that follows the body's form; do not answer 'indeterminate' merely because little or no skin is visible, as long as the body's outline and shape are visible. Do not answer 'indeterminate' for any other reason, and do not answer 'indeterminate' merely because you are not fully certain – if the body's build is genuinely visible, give your best yes/no judgment. Answer with exactly one word: yes, no, or indeterminate."

Visible presentation, not underlying body composition. The preregistered criterion operationalises thinness as *visible* body fat, rather than an anthropometric measure (e.g., Body Mass Index). Classification therefore reflects the body presentation depicted in a given still, not necessarily the depicted individual's actual body composition; a still in which a person's build presents as thin due to pose, angle, or lighting is correctly classified as thin under this criterion, consistent with the study's focus on exposure to thin-presenting content rather than on the true body composition of individuals appearing in it.

Response length and refusal handling. As in Step 5, the prompt constrains the model to a maximum output of five tokens, comfortably sufficient for the required single-word response but insufficient to produce extended reasoning; this reflects the same deliberate design rationale described for Step 5, that providing room for justification tended to make the model more willing to hedge rather than improving judgment quality. Responses are additionally screened for model refusal (the model declining to engage with the judgment at all), tracked as a distinct *refused* category separate from indeterminate. This screening was considered particularly relevant for thinness relative to Step 5, since a judgment about an individual's body build is plausibly more likely than a clothing or posing judgment to trigger a model's own trained caution. *Results.* Of the 1,050,974 person-crops,

565,675 (53.82%) were classified as showing a thin body build, 83,655 (7.96%) as not, and 401,627 (38.21%) as indeterminate; the model refused to classify 17 (<0.01%). Combining the three classifications, 32,441 person-crops (3.09% of the Step 4b pool) satisfied the confirmatory definition used in the paper (minimal/tight clothing, posing of any kind, and thin body build; Section 3.7), and 4,729 (0.45%) satisfied the preregistered definition (minimal/tight clothing, sexually suggestive posing, and thin body build). The confirmatory crops fall in 30,842 stills, i.e. 3.64% of the 847,714 stills with at least one classified crop and 0.94% of all 3,283,135 stills on screen.

3.6.1 Step 6 – Manual validation by independent reviewers.

We conducted a blinded manual validation of 1,000 randomly sampled frames using two independent human reviewers. On average, the reviewers agreed with the visual model on 71.45% of frames, with substantial interrater reliability between the reviewers (Cohen's $\kappa = 0.66$; 86.90% agreement).

3.7 Step 7 – From classified stills to exposure seconds: preregistered versus confirmatory definition

Steps 1–6 yield a classification for every person-crop. Exposure is computed at the level of the still: a still qualifies if at least one person-crop in it meets the definition, and each qualifying still contributes its on-screen time (one second for video frames, three seconds for slideshow images; Section 6.1) up to the estimated watch duration of the video (Section 6.4). Daily exposure is the sum of these seconds across all stills a participant viewed on a given day, with substitution for videos that could not be classified (Section 6.5).

Preregistered definition. The preregistration defined sexualised content as the conjunction of minimal/tight clothing, sexually suggestive posing, and thin body build. Under this definition, 4,729 person-crops qualify (Step 6; 4,712 stills). Sexually suggestive posing was rare in the content participants were served: 11,702 of the 1,050,974 person-crops entering VLM classification (1.11%), or approximately 0.2% of all 5,568,853 detected persons, against 618,987 (58.90%) showing deliberate posing of any kind. At the exposure level this scarcity is decisive: under the preregistered definition, 31.4% of donation-covered participant-days had zero exposure, and the median daily exposure was 3.3 s (IQR 0.0–11.1). A predictor concentrated at zero cannot support a within-person test, since there is almost no day-to-day variation for the model to use.

Modified definition. The analyses therefore retain posing of any kind (Step 5b, Stage 1) as the posing component, replacing sexually suggestive posing (Stage 2) with a broader operationalisation of the same construct dimension: deliberate, camera-directed self-presentation. The definition used throughout the paper is minimal/tight clothing, posing, and thin body build. Under it, 32,441 person-crops qualify (30,842 stills), 84.5% of participant-days had nonzero exposure, and the median daily exposure was 28.8 s (interquartile range 5.5–67.1). Table S9 compares the two definitions at each level.

Sensitivity analysis under the preregistered definition. In the complete-case model (n = 740 participant-days from 66 participants; the registered model with random intercept, random within-person slope and AR(1) residuals), daily exposure to thin and sexualised female bodies under the preregistered definition did not predict end-of-day body satisfaction within persons (H1a: β_within = −0.109 per minute, SE = 0.145, 95% CI [−0.393, 0.174], one-sided p = .225) or between persons (H3a: β_between = 0.415, SE = 0.946, 95% CI [−1.474, 2.304], one-sided p = .669); neither coefficient reached the preregistered significance level (one-sided α = .05),

matching the pattern of the confirmatory analyses. The multiply imputed version of this model was not run.

Table S9. *Prevalence of qualifying content under the preregistered and confirmatory exposure definitions.*

Level	Preregistered (clothing/ suggestive posing/ thin)	Modified (clothing/ any posing/ thin)
Person-crops qualifying (of 1,050,974 entering VLM classification)	4,729 (0.45%)	32,441 (3.09%)
Stills qualifying (of 847,714 stills with at least one classified crop)	4,712 (0.56%)	30,842 (3.64%)
Total exposure-seconds, classified videos (of 3,487,663 presentation seconds on screen)	4,888 (0.14%)	31,803 (0.91%)
Participant-days with nonzero exposure	623 of 908 (68.6%)	767 of 908 (84.5%)
Median daily exposure, s (IQR)	3.3 (0.0–11.1)	28.8 (5.5–67.1)

3.8 Video processing and pipeline validation

A total of 242,422 videos were processed using YOLO26 person detection, yielding 5,568,853 detected person instances (23.0 per video on average). Each detected person was evaluated with a pose-based visibility check requiring at least partial visibility of the face, torso, or thighs; 4,710,317 instances (84.6%) met this criterion and entered gender-presentation classification. The

female-presence filter (Section 3) classified 3,167,378 persons (56.9%) as female-presenting; excluding persons whose only visible region was the face retained 2,892,484 person-crops, contained in 1,912,733 of 2,880,586 stills (66.4%), for subsequent screening. Complete stage-by-stage counts through the clothing, posing, and thinness classifications – culminating in 32,441 person-crops satisfying all components of the confirmatory definition (4,729 under the preregistered definition) – are reported in the Supplementary Material, Section 3.

Manual validation of the pipeline by independent, blinded reviewers is summarised in Section 3.9 and in the main text: agreement with the classifier was 94.4% (human presence), 89.5% (female presence), 78.5% (clothing pre-filter), 71.3% (sexualised presentation) and 71.5% (thinness).3

4 Exploratory prompt-level analysis: specification and full results

The exploratory prompt-level analysis reported in the main text used windows running from one answered prompt to the next within a day: from the morning prompt to the afternoon prompt, and from the afternoon prompt to the evening prompt. Windows across the night were not used. This gave 1,242 windows from all 66 participants, 1,221 of which fell within the donation period. Exposure within a window was computed in the same way as daily exposure in the main analyses. The model had a random intercept and a random exposure slope per participant and continuous-time AR(1) residuals, and it adjusted for body satisfaction at the previous prompt. Higher-than-usual exposure in a window was not associated with body satisfaction at the prompt that followed ($\beta = -0.011$ per within-person SD of the exposure rate, SE = 0.024, 95% CI [−0.059, 0.036], one-sided $p = .318$).

5 Social media data donation

Under the UK General Data Protection Regulation (Article 15, the right of access), as implemented in the Data Protection Act 2018, any individual has the right to request and receive an electronic copy of their personal data from any data controller. Most major platforms implement this right by allowing users to request "data download packages": machine-readable electronic files containing the personal data the platform holds about the requesting user. Data donation research [15] builds on this right by asking participants to request such a package and voluntarily share it, in whole or in part, with a research team. Because the packages are retrospective, a single donation at the end of the study period suffices to reconstruct exposure across that period.

The project focuses on TikTok for two reasons. First, TikTok is consistently among the platforms most widely used by UK adolescents, alongside YouTube and Instagram.[16] Second, at the time that the study was conducted TikTok was among the only major platforms whose data download packages contain time-stamped watch-history records identifying the specific content viewed, [17] enabling reconstruction of users' real-time content exposure. To minimise participant burden while maximising scientific value, the content-exposure measurement in this study was therefore limited to TikTok.

6 Content timing, watch-duration estimation, and unclassifiable content

This section gathers every timing decision in one place: how long each still was on screen, how long the content itself lasts, how long each viewing event was watched, and how content that could not be classified was treated.

6.1 How long each still was on screen

A video still represents one second of playback, because frames were extracted at exactly one per second. A slideshow image is credited three seconds, which is TikTok's photo-post advancement rate.

Because the three-second interval determines both the weight of every slideshow still and how many stills a given watch duration reaches, it was verified against the platform rather than taken from documentation, through three manual checks (July–August 2026). First, the advance was timed directly in the platform application on slideshow posts of differing length and audio type; in every case the displayed image advanced at three seconds. Second, short slideshows were observed to cycle through their images repeatedly rather than stopping after a single pass: a five-image post with a sixty-second soundtrack, for example, ran four complete cycles and then stopped, so a slideshow's total playback is generally a multiple of three seconds times its number of images rather than equal to it. Third, posts at the extremes of the audio-to-image-count ratio described below were opened and inspected individually.

We had initially intended to corroborate the interval, and to obtain each slideshow's playback duration, from the donated files, by measuring each post's audio track and comparing it with the post's number of images. This proved unreliable for both purposes. Across the 14,558 slideshows with 8–15 images the implied seconds-per-image ratio was a featureless distribution (25th

percentile 2.0 s, median 3.6 s, 75th percentile 5.5 s) with no modal value – the signature of two unrelated quantities rather than of display behaviour. Inspecting the extremes identified the two mechanisms: at the low end, posts carrying a brief audio cue on each image rather than a continuous soundtrack (3–6 seconds of audio on posts of 10–14 images, each displayed for three seconds); at the high end, posts with a complete attached music track of several minutes, which does not play to its end. Only where the soundtrack genuinely spans the post does the ratio recover the display interval (2.95–3.03 s across a sample of such posts). Audio duration is therefore not used anywhere in this pipeline.

6.2 How long the content itself lasts

A post's content duration is derived from the stills the pipeline holds: for video, the number of extracted stills (one per second); for slideshows, three seconds times the number of images, that is, one complete pass. No separately measured media length is used.

For video this was validated against the durations measured from the downloaded files during an earlier processing stage. Across the 319,140 videos holding both a stills folder and a measured length, the still count matched the measured length to within one second for 99.973%, with an offset of zero or one frame (the frame at t = 0). All 75 disagreements ran in the same direction, with fewer stills than measured seconds; inspection showed these to be videos whose visual content ends before the clip does, typically a static final image held to the end. For those videos the still count is the duration of *distinct content*, which is the appropriate bound on exposure, whereas the measured length would credit watch time against frames that were never classified.

6.3 Why each still is counted once

A viewing event can last longer than the content it shows, and in both content types the mechanism is a loop. A TikTok video replays from the start when it reaches its end and the viewer has not moved on; a slideshow cycles through its images repeatedly until playback ends (Section 6, first subsection). In the watch-duration cascade this situation is visible directly: whenever the session-aware gap to the next event exceeds the content's own length – the "full length used" branch, 93,090 unique watch events (20.8%) – the participant remained on the post for longer than one pass, and the content looped for the remainder. For slideshows the same applies wherever the gap exceeds three seconds times the number of images, which for the short slideshows that dominate the corpus (median three images) is the common case.

Exposure is defined here over *distinct* stills: each still contributes its on-screen duration at most once per viewing event, so viewing time beyond one complete pass contributes no additional exposure. A twenty-second view of a three-image slideshow thus contributes nine seconds across three distinct images rather than twenty seconds across repeated images, and a thirty-second view of a fifteen-second video contributes that video's fifteen stills once each. In the cascade this is what the content length does: it caps each event's watch time at one pass, and the "capped by the gap" branch (259,166 events, 58.0%) covers the events where the viewer left before the end and no loop occurred.

This is a definitional choice rather than a preregistered rule, and the alternative – crediting repeated passes cumulatively, on the reasoning that dose is time on content – is defensible. Counting distinct stills was adopted for three reasons. First, whether seeing the same image again immediately after the first constitutes a second exposure of equivalent effect is not established; repeated presentation of an identical stimulus may be attenuated by habituation rather than accumulating linearly, and

we are not aware of evidence that would settle the question for this content and this outcome. Counting each still once is the conservative option under that uncertainty, since it cannot inflate exposure through a repetition effect that may not exist. Second, the rule must apply to both content types: video loops and slideshow cycles arise from the same gap-exceeds-length condition, so crediting repetition for one would require crediting it for the other, and any asymmetry would itself need justification. Third, the construct of interest is the composition of the content a participant saw rather than cumulative viewing time.

6.4 How long each viewing event was watched

Watch histories record when a video appeared, not how long it was watched. Each of the 446,709 unique watch events contributed by the 66 participants within their assessment windows therefore received an estimated duration through the evidence-ranked cascade described above.

The cascade first constructs a session-aware gap for every event, using the donated session logs to distinguish a gap that falls within a viewing session from one that spans a break in use; where a break was crossed, the time to the session's own end replaces the raw gap. Where the content's own length is known (352,256 events, 78.9%), watch time is the shorter of that length and the gap: the gap bound applies in 259,166 events (58.0%), where the viewer moved on before the content ended, and the full length in 93,090 (20.8%). Where no content length is available (94,453 events, 21.1%, all of them posts with no images on disk), the gap itself becomes the watch time, and how well the gap is evidenced determines the route: a session-confirmed gap in 85,759 events (19.2%), the session's end time in 1,588 (0.4%), and a gap no longer than the participant's own typical viewing duration in 4,786 (1.1%). In the remaining 2,320 events (0.5%) no usable bound existed

and a duration was drawn at random from that participant's own distribution of high-confidence durations (hot-deck substitution, fixed seed).

6.5 Content that could not be classified

Of the 314,517 distinct videos the 66 participants watched within their assessment windows, 211,943 (67.4%) could be retrieved and classified, accounting for 319,161 unique watch events (71.5%) and 72.8% of estimated watch time. The remaining 102,574 videos (32.6%; 127,548 unique watch events, 27.2% of watch time) carry no content labels. Almost all of these (102,532) could not be retrieved at all; for 42, media were partially obtained (an audio file exists) but no images, which is a fact about the download rather than about the post, since every TikTok post has visual content.

Both groups are treated identically and are neither dropped nor coded as containing no idealised content. Their watch time is retained in the daily totals – the participant did spend that time on TikTok – and their content composition is estimated by the preregistered content-probability substitution (Section 2), which assigns them the qualifying proportion observed in that participant's classified viewing on the same day, falling back to her own overall proportion and then to the sample proportion. This treats unclassifiable content as content of unknown composition rather than as an absence of exposure. Coding it as zero would assert something about content that was never observed; removing it from the totals would shrink the denominator and inflate every exposure proportion.

6.6 Viewing-history processing

This section describes how participants' TikTok viewing histories were processed and matched to classified content data to compute daily exposure, uniformly across all three exposure

operationalisations (thin-and-sexualised body content, beauty-filter use, beauty-labelled content), since all three require the same underlying estimate of how long each donated video was actually watched.

6.6.1 Video duration estimation

For each TikTok video a participant watched, an estimate of watch duration was required. Where the video's own, directly-measured duration was available, this was used directly, unless direct behavioural evidence indicated the participant switched away before the video finished. Across the 66 included participants, donated watch histories comprised 446,709 viewing events, of which 350,521 (78.5%) had a directly-measured duration (81,977 used unmodified; 268,544 capped by a shorter subsequent gap). Where a video's own duration was known, the elapsed time until the participant's next recorded TikTok viewing event provides direct behavioural evidence of how long that specific video was actually watched, whenever that gap was shorter than the video's own full length; whenever the gap to the next viewing event was both finite and shorter than the video's known duration, the shorter, better-evidenced gap value was used in place of the full duration. A video's own known duration was used unmodified only when no such shorter gap existed to reduce it.

6.6.2 Session-aware gap estimation

Session boundaries were derived directly from session-level data included in participants' donated data packages (start time and duration of each discrete TikTok use session), rather than inferred from viewing timestamps alone. Where a participant's subsequent viewing event fell after their current session had already ended, the elapsed time until that session's own end – rather than the elapsed time until the next recorded viewing event, which can additionally include time spent

entirely outside the TikTok app – was used as the more precise estimate of engagement with the current video.

6.6.3 Videos with no directly-measured duration

For viewing events without a directly-measured duration (96,188 events, 21.5%), the elapsed time to the next viewing event was used as a duration estimate in its own right only when session data confirmed that subsequent event occurred within the same continuous session. Where session data instead confirmed a session boundary was crossed before the next viewing event, the elapsed gap was not used as a positive duration estimate; the elapsed time until the current session's own end was used instead. For a smaller subset of videos with no directly-measured duration and no session data available, plausibility was instead assessed against each participant's own typical (median) directly-measured video duration; gaps shorter than or equal to a participant's own typical duration were used directly, and gaps exceeding it were treated as unreliable.

6.6.4 Substitute values

Where no duration estimate could be derived from a video's own measurement or the gap to a subsequent viewing event, a substitute value was required. Such cases were rare: 2,284 events, or 0.51% of all viewing events. Each substitute value was drawn at random from that participant's own set of directly-measured, same-session-confirmed video durations – a single stochastic (hot-deck) imputation in which the donor pool is the participant's own observed duration distribution, chosen to preserve day-to-day variability in video length rather than collapsing it to a single repeated value. Where a participant had no such durations of their own, the substitute was instead drawn from the pooled set of all participants' such durations. Table S11 summarises the source of every viewing event's watch-duration estimate.

Table S11. *Source of each viewing event's watch-duration estimate (N = 446,709 viewing events across 66 participants).*

Source	N	%
Directly measured duration, used unmodified	81,977	18.4
Directly measured duration, capped by a shorter subsequent gap	268,544	60.1
Gap to next viewing event used directly (same session confirmed)	87,362	19.6
Gap used directly (no session data; within participant's typical duration)	4,925	1.1
Session-end-based duration (session boundary crossed)	1,617	0.4
Random draw substituted (no subsequent same-session viewing event)	12	<0.1
Random draw substituted (no session data; gap exceeded typical duration)	2,272	0.5
Total	446,709	100.0

6.6.5 Extraction of viewed stills

Videos were segmented into still images at one still per second of video content, or one still per 3 seconds of slideshow content (matching the slideshow's own advancement rate). Per the preregistration, only stills up to each video's estimated watch duration were extracted for content classification, regardless of the video or slideshow's total available length.

6.6.6 Content-probability substitution for missing videos

For videos successfully downloaded and classified, daily exposure was computed by summing the presentation duration of all stills classified as meeting the relevant criterion on a given participant-day. For videos a participant watched but that could not be downloaded (and therefore could not be classified), the two-tier content-probability substitution procedure described in Section 6.5 was

applied: each missing video's substituted exposure contribution was calculated as its estimated duration multiplied by the participant's own day-level proportion of classified exposure among their other, available videos that same day (tier 1), falling back to their own proportion across their full donation period only when no videos were classified for them on that day at all (tier 2). This procedure was applied to all donation-covered participant-days in the analysis grid (908 of 924); the remaining 16 participant-days fell outside donation coverage and carry no derivable exposure. Nearly all donation-covered days required content-probability substitution for at least one video. In terms of total exposure time, 29.0% of total exposure-seconds across the dataset were attributable to substituted, rather than directly-classified, videos for the thin-and-sexualised exposure measure.

7 Beauty-filter reference list

7.1 Source data

For each video in a participant's watch history, the TikTok metadata list the sticker effects applied to it, each with a name and an effect ID. Across the 813,515 videos with metadata, 68,947 (8.5%) carried at least one effect, and 5,281 distinct effect names occurred. Green-screen effects, which replace the background rather than alter the person, were the most frequent effects in the data (e.g., "Green Screen", 6,672 videos) and were set aside at the outset.

7.2 Working definition

Following the preregistration, a beauty filter is a sticker effect that alters a user's facial or body appearance to conform more closely to idealised beauty standards — skin smoothing, facial reshaping, makeup application, eye enlargement, lip enhancement, hair modification, or other cosmetic enhancements.[18] Effects that alter the face without beautifying it (comic distortions, character effects, deliberately unflattering looks), camera and display effects (flash, fisheye, zoom, mirroring), interactive games, quizzes and rankings, and overlays unrelated to appearance do not meet the definition.

7.3 Preregistration stage: candidate list

The candidate list was compiled before the analysis and preregistered (OSF : https://osf.io/chbxw). Every distinct effect name present in the data was considered. MZ used Claude Opus 5.5 to develop and refine a set of appearance-related keywords intended to capture multiple dimensions of appearance modification. These included facial features and body characteristics (e.g., "eye," "nose," "teeth," and "slim"), cosmetic products and techniques (e.g., "makeup," "contour,"

"blush," and "foundation"), and broader aesthetic descriptors (e.g., "beauty," "pretty," and "glow"). The resulting keyword set comprised: ["face," "facial," "eye," "lash," "hair," "lip," "brow," "teeth," "nose," "chin," "jaw," "forehead," "skin," "makeup," "blush," "contour," "foundation," "highlight," "glam," "blond," "ginger," "pretty," "smile," "slim," "smooth," "tan," "glow," "clear," "beauty," and "flawless"]. The keywords were intended to identify a broad range of potentially relevant effects rather than constitute an exhaustive or mutually exclusive taxonomy. An effect name containing any keyword was flagged as a candidate for inclusion. Because some keywords were deliberately broad (e.g., "face"), this step favoured recall over precision.

Second, MZ manually examined effect names present in the data and added appearance-altering effects whose names were not captured in keyword matching (e.g., "prettiest", "cutest"). The union of the two steps formed the preregistered candidate list of 489 effect names. This list defined which effects were eligible for the beauty-filter measure; it was compiled from effect names only, without viewing the videos.

7.4 Analysis stage: verification review

Because the candidate list was built from names, it could include effects that match a keyword but do not beautify. Before computing exposure, CH therefore reviewed every one of the 484 candidate names against the working definition in Section 7.2.

Names that made the effect's action clear were classified directly. Effects whose names stated a beautifying action were retained (e.g., "Smooth Skin by Sophie", "Teeth whitening", "Natural Makeup", "Sharp jawline"); colour effects were retained only where the name indicated a natural-looking hair or eye colour (e.g., "Blonde Hair", "Hazel Eyes"). Excluded by name were effects that alter the face or body without beautifying it – comic distortions and exaggerations (e.g., "Ugly

Face", "Squash Face", "Stretchy Face", "Big Nose", "Big Forehead"), character and animal looks (e.g., "Clown Face", "Gorilla Face", "Frog Face", "Ed Sheeran Face"), deliberately unflattering looks (e.g., "ACNE SKIN", "eyebags", "Yellow Teeth Test", "Bad Haircut", "Fail Makeup", "Spooky SFX Makeup") and novelty colours and accessories (e.g., "Glowing White Eyes", "Heart Eyes", "rainbow hair by jrm", "Piercings") – as well as effects unrelated to appearance: camera and display effects (e.g., "Rotating Fish Eye Lens", "Front Flashlight", "Face Zoom", "Flip Reverse Mirror", "Pixel Face"), quizzes, games and rankings (e.g., "Eye Test – 1", "Hair Ranking", "Lip Gloss Ranking", "What Job Suits your face", "Gojo Clash Battle", "Filipino Food Rank") and overlays (e.g., "#ISTAND-WITH-PALESTINE", "pink bow", "glasses").

Where the name left the effect's action unclear, the effect was looked up on TikTok, whose discover pages list public videos by the effect applied to them (e.g., tiktok.com/discover/rosmar-tan-filter), and its visible action was inspected there; where several effects share a name on the platform, the look-up reflects the effects returned under that name. Effects were not inspected in the videos of our own data.

The review retained 243 of the 484 candidate names and excluded 241. The final reference list thus contains 243 effect names, listed in Table S12 with a high-level categorisation of what each effect does.

7.5 Classification of videos

A video was classified as beauty-filtered if at least one of its effects was on the final reference list, matched by name (case-insensitive, after trimming surrounding spaces). Of the 318,130 videos watched on the study days, 227,335 (71.5%) had a metadata row; of these, 17,107 (7.5%) carried at least one effect and 5,463 (2.40%) were beauty-filtered under the final list, accounting for 1.79%

of the watched seconds in videos with metadata. Videos without metadata (90,795; 28.5%) were treated as unclassified due to being unavailable at the time of the analysis and handled by the substitution procedure in Section 6.5.

7.6 Deviation from the preregistration

The preregistration described the reference list as constructed a priori and did not foresee a verification step. The verification review is a post-registration refinement of the operationalisation, applied uniformly to all eligible effects before any model was fitted; it narrows the list but does not change the definition.

Table S12. *Beauty-filter reference list: the effects included after the verification review, grouped by what the effect does. Names are reproduced exactly as they appear in the metadata, including capitalisation and creator tags; within a group they are in alphabetical order.*

What the effects do	Effect names (as written in the TikTok metadata)
makeup (foundation, blush, contour, eyes or lips) (97)	2000s Makeup; 2016 Makeup by Paige; 90s Pam Makeup; Amazing eyeliner; Angelina Makeup; barbie bratz contour; Barbie contour; Beauty Lashes Lipstick; BECA (makeup); Big Gloss; Big Lips; Blush; Blush Makeup; blush tones; Bold Glamour; Brown makeup; butter blush; cat eye; Cinnamon Girl Makeup; clean girl make up; Contour face; Contouring by Maytte; Cool Pink Lips; cute eyeliner; Cute Makeup; Cute Makeup x Paige; Daily Makeup; dot contour; Eternal Sunshine Makeup; Extreme Smokey Eye; Eyeliner; Face contour; facelift contour; Fall Makeup; falsies n highlighter; glam by jrm; glitter lip; Gold Color Highlighter; grape makeup; Hotness x Makeup; i love u highlight; just makeup; KOREAN MAKEUP; Lady Sapphire Makeup; Lashes and Makeup; latina makeup; Latina Makeup; light makeup; Lip Makeup; lip pink 9; lip xinh 8; Lips by "Shelove"; lips by aramkk5; Lite Foundation; make up mèo 5; makeup; makeup baruu; makeup x arda; Makeup-katelynne; Mob Wife Makeup by Paige; model contour; My MakeUp - @kirayle; Natural Makeup; natural makeup; New makeup II; no make up baby; no makeup baby; No Makeup by julisadg; no rule contour; nose contour; Ombre Lips; pink makeup; puppy eyeliner; Red Lips; Sapphire S9xy Makeup; Shakira Makeup by Paige; Shine Lady Makeup; Silki Makeup Lady; simple makeup; SImple Makeup; Siren Eyes by Sophie; Soft Blush; Soft glam; SOFT MAKEUP; soft makeup; Spring Eyemakeup II; Strawberry + Blue Eyes; Strawberry + Hazel Eyes; Strawberry makeup; strawberry makeup; Strawberry Makeup; Strawberry makeup RB; Subtle Eye Glam by Sophie; Subtle Makeup by Sophie; Tinted Lip Balm by M2M; Top Sapphire Makeup; WATERMELON BLUSH
eyelashes (32)	5D Eyelashes; Bambi lashes; cute freckles with lashes; cute lashes; extra lashes; Eyelash; Eyelashes; eyelashes; EYELASHES 2x; Eyelashes 6D; Eyelashes A; Eyelashes by "Shelove"; Eyelashes Line Lady; Fluffy Lashes; freckles and eyelashes; frecks and lashes; indie lash by emmasofija; just lashes; Kylie lashes; Lash; lashes; lashes with hazel eyes; manga eyelashes; Manga Lashes; natural eyelash; only soft lashes; Perfect lashes; Perfect Lashes; PRETTY BABE x LASHES; real freckles with lashes; Simple Lashes; THICK eyelashes

What the effects do	Effect names (as written in the TikTok metadata)
eye colour or eye enlargement (17)	Big Eye Baby; big eyes; Big Round Cat Eyes; Black hair and blue eyes; Blue Eyes; Blue Eyes Shiny; brown doe eyes; Brown Eyes; Brown eyes; Eye Colors; Green eye 2024; green eyes; Green Eyes; Green Eyes for Brown Eyes; Hazel Eyes; MY BLUE EYES; Only blue eyes
skin smoothing / glow (14)	CLEAR SKIN; Cute Glow; Flawless Skin; Gentle Glow; glowing; LO GLOW by julisadg; Natural Glow; natural skin; Natural teen face; perfect skin; prettiest; PRETTY BABE; Smooth Skin by Sophie; Smooth Soft Skin
face or feature reshaping (5)	Button Nose; Just baby slim; No filter slim; Sharp jawline; straight nose
teeth whitening (3)	Teeth Brightener; Teeth whitening; Whitening Teeth
tan / skin tone (9)	light tan; Natural but tanned; Rosmar Tan; TAN; tan effect; Tan Effect; Tanned Cute; tanned cutie; Tanning Makeup
freckles (16)	Faux Freckle; Faux Freckles; Freckles; freckles by emmasofija; freckles by emmasofija 4; Freckles by Sophie; freckles KID RB; freckles without catfish; freckless; Ginger Freckles; Ginger hair and freckles; Good Freckles; Green eyes freckles; Natural Freckles; real freckles; realistic freckles
hair colour (13)	Blonde Hair; Blonde Hair Color; Blondy; Brown Hair; Brown Hair by julisadg; Brown Hair II; cherry hair by aramkk5; cold blonde hair; Cowboy Copper for Blondes; ginger by jrm; Red Hair Filter; silky hair; strawberry blonde by jrm
overall beautifying look (smoothing, reshaping, light makeup) (37)	aesthetic beauty; ALLURE by mayttearaneda; angel face; baby babe; Baby Doll; beauty; Beauty Plus; beautyeffect; Chic Cutie; Crystal Bratz; cute by emmasofija; cutest by Paige; cutie; doll; fine face; Flawless beauty YF; flawless look; Glamour audace; Hazel Face by Maytte; Lite Perfect 😍; My Beauty; My Beauty YF; my perfect face; natural; Natural; Natural beauty TE78; Natural plain face XR75; No Filter; No filter 3; NO FILTER FILTER; No filter sweet lips; Princess; Slavic Princess; SOFT BEAUTY PLUS; Striking Face; Teenage Look; You are cute

8 Stability of body satisfaction over the study period

Figure S2 shows every participant's evening body-satisfaction ratings across the 14 study days, the picture behind the intraclass correlation and the within-person variation reported in the Results.

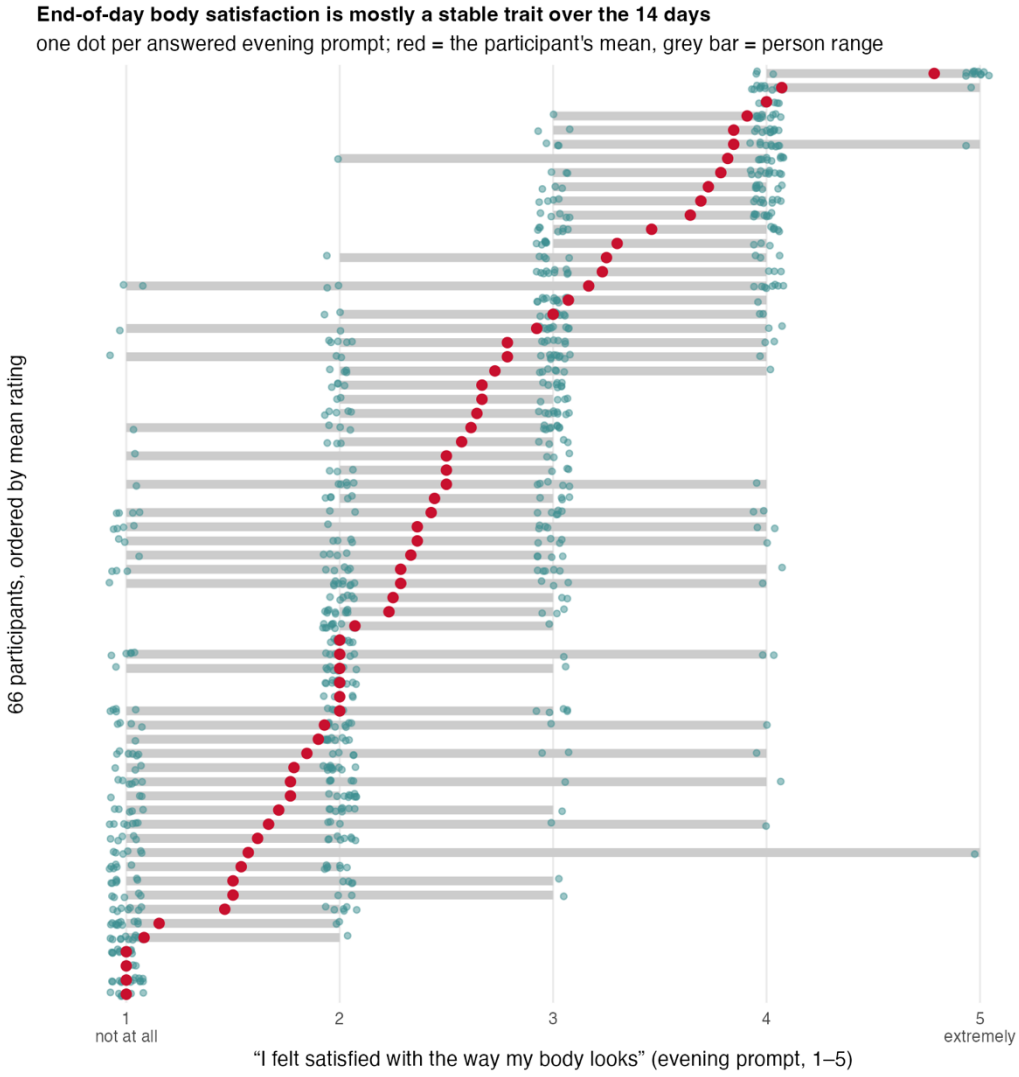


Figure S2. *Stability of end-of-day body satisfaction over the 14 study days. One row per participant (n = 66), ordered by individual mean ratings; one dot per answered evening prompt (up to 14); the red point is the person mean and the grey bar a person's individual range on the 1–5 item "I felt satisfied with the way my body looks".*

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